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An Edge-Cloud Sensor-Agnostic Framework for Real-Time Microseismic Monitoring and Analysis

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Abstract

Induced seismicity is a critical diagnostic tool for monitoring subsurface processes in a variety of energy applications, including carbon capture and storage (CCS), hydraulic fracturing, geothermal energy, and enhanced oil recovery. In the context of CO₂ injection, microseismic monitoring enables the detection of acoustic emissions generated by rock failure, providing real-time insights into reservoir dynamics, plume migration, and the integrity of both the reservoir and cap rock. These microseismic events serve as sensitive indicators of stress changes near the wellbore and within the reservoir, offering a means to identify and characterize faults or fracture zones that may be activated during injection. This information is essential for assessing containment risks and constraining the maximum expected magnitude of induced seismic events.

This paper presents a comprehensive workflow for passive seismic monitoring using Distributed Acoustic Sensing (DAS), a fiber-optic-based technology that is gaining traction in the oil and gas industry due to its deployment flexibility, cost-effectiveness, and ability to transform existing fiber infrastructure into dense seismic arrays. DAS enables high-resolution, large-aperture monitoring systems that can be deployed temporarily via wireline or permanently as part of well completions. Building on extensive experience from 3C sensor-based microseismic campaigns, we propose a DAS-centric monitoring framework designed to enhance event detection, characterization, and long-term monitoring capabilities—particularly in support of Measurement, Monitoring, and Verification (MMV) programs.

The workflow begins with the optimization of monitoring geometry through the evaluation of event detectability and location uncertainty across various sensor configurations. A detailed velocity model is developed, incorporating geological, geophysical, and geomechanical parameters, including rock attenuation and ambient noise characteristics. Where available, geomechanical data are used to model expected source mechanisms and avoid nodal planes, improving the reliability of event detection.

Waveform simulations for various failure mechanisms are used to benchmark expected seismic signatures. A real-time DAS data simulator is employed to validate the acquisition geometry and processing pipeline under realistic operational conditions. To automate event detection, we introduce a novel real-time processing framework that combines traditional geophysical techniques with image processing algorithms, emulating expert interpretation of DAS-acquired data.

The integrated workflow supports both conventional 3C and fiber-optic-based systems, reducing uncertainties in event localization and source mechanism analysis. This workflow is effective in monitoring CO₂ injection wells over extended periods, ensuring safe operations and regulatory compliance. The methodology is equally applicable to hydraulic fracturing, geothermal energy production, and long-term passive monitoring in oil and gas fields, offering a robust, scalable solution for real-time subsurface surveillance.

Introduction

Induced seismicity plays a critical role in subsurface monitoring applications across a range of energy sectors, including hydraulic fracture stimulation monitoring (Warpinski et al., 1999; Maxwell and Urbancic, 2001; Le Calvez et al., 2005; Warpinski, 2009; Van Der Baan et al., 2013; Warpinski and Wolhart, 2016), geothermal energy (Dyer et al., 2008; Julian et al., 2009, Verliac and Le Calvez, 2021; Grigoli et al., 2022), carbon storage (Martinez-Garzón et al., 2013; Smith and Jacques, 2016; Le Calvez et al., 2021; Li et al., 2022). It offers a valuable means of tracking CO₂ plume migration and assessing the integrity of both the reservoir and cap rock during carbon storage operations. Microseismic monitoring detects acoustic emissions generated by rock failure, often triggered by fluid injection, such as stimulation fluids, CO₂, into subsurface formations. These microseismic events serve as sensitive indicators of stress changes near the wellbore and within the reservoir. Once injection begins, real-time seismic monitoring becomes essential for understanding reservoir dynamics and ensuring containment integrity.

This study presents a comprehensive workflow for passive seismic monitoring using Distributed Acoustic Sensing (DAS), a technology that is gaining traction due to its deployment flexibility and ability to transform fiber optic cables into dense seismic arrays. Building on insights from numerous 3C sensor-based monitoring campaigns, we propose a DAS-centric approach that enhances event detection and characterization capabilities, making it well-suited for long-term monitoring and Measurement, Monitoring, and Verification (MMV) programs.

The workflow begins with the optimization of monitoring geometry through assessments of event detectability and location uncertainty, using a detailed velocity model that incorporates geological, geophysical, and geomechanical parameters. Next, we simulate waveforms for various failure mechanisms to benchmark expected seismic signatures. A real-time DAS data simulator is then employed to validate the acquisition geometry and processing pipeline under realistic conditions.

To automate event detection, we introduce a novel real-time processing framework that combines traditional geophysical techniques with, when appropriate, image processing algorithms, emulating expert interpretation of DAS-acquired data.

This integrated workflow enables reducing uncertainties in event localization and source mechanism analysis. Field deployments have demonstrated its effectiveness in monitoring CO₂ injection wells over extended periods, enabling safe, real-time operations with minimal human intervention.

Background

Microseismic monitoring has long been a key technique for understanding subsurface processes in energy-related operations. It involves detecting and analyzing small-scale seismic events—typically with moment magnitudes less than 0—that occur due to stress changes in the rock mass. These events are particularly useful for mapping fracture networks, identifying fault activation, and assessing the mechanical response of the reservoir to fluid injection or extraction. In carbon capture and storage (CCS), for instance, microseismicity can reveal the migration path of injected CO₂ and provide early warning of potential containment breaches. Similarly, in hydraulic fracturing, it helps delineate stimulated rock volume and monitor fracture propagation.

Traditional microseismic systems rely on arrays of 3C geophones or accelerometers deployed in boreholes or on the surface. While these systems offer high-fidelity data, they are often limited by high deployment costs, logistical complexity, and sparse spatial coverage. Additionally, their fixed sensor spacing can restrict resolution, especially in large or complex reservoirs.

Distributed Acoustic Sensing (DAS) has emerged as a disruptive alternative. DAS systems convert standard fiber optic cables into dense arrays of seismic sensors by measuring strain-induced changes in backscattered light. This allows for thousands of sensing points along a single fiber, enabling high-resolution seismic imaging over long distances. DAS can be deployed using existing downhole or surface fiber infrastructure, significantly reducing installation costs and improving operational flexibility.

Despite its advantages, DAS presents unique challenges, including directional sensitivity, lower signal-to-noise ratios compared to 3C sensors, and the need for specialized processing techniques. However, recent advances in DAS hardware, signal processing, and machine learning have significantly improved its performance and reliability.

This paper builds on these developments by proposing a DAS-centric workflow that integrates best practices from traditional microseismic monitoring with innovations tailored to the capabilities and limitations of DAS technology.

Workflow Architecture

The proposed workflow is designed to leverage the strengths of Distributed Acoustic Sensing (DAS) for passive seismic monitoring while addressing its unique challenges (Mizuno and Le Calvez, 2024a). It integrates key elements from traditional 3C microseismic workflows with innovations tailored to DAS data acquisition and processing (Le Calvez et al., 2022). The architecture is modular, scalable, and adaptable to a variety of subsurface monitoring scenarios, including CO₂ injection, hydraulic fracturing, geothermal energy production, and long-term reservoir surveillance.

The overall product has two main components (Mizuno and Le Calvez, 2024a):

- Edge component which performs the real-time automatic event detection, location and magnitude computation. ‘Edge’ refers to the computing location i.e., on-site, in proximity to the well/sensor(s) deployed for monitoring. The Edge component consists of mainly the DAS interrogator and the processing computer (e.g., a high-performance laptop for example) deployer on site or available in a nearby unit such as a wireline one.
- Cloud component which provides the real-time monitoring and the remote quality check both at any chosen connected location.

At its core, the workflow consists of four interconnected components: (1) monitoring geometry optimization, (2) seismic waveform simulation, (3) real-time DAS data simulation, (4) automated event detection and classification. Each component is designed to operate independently or as part of a continuous monitoring loop, enabling both pre-operational planning and real-time decision support.

The workflow begins with the design and optimization of the sensor array layout, using synthetic modeling to assess event detectability and location uncertainty. This step ensures that the DAS system is configured to maximize sensitivity and spatial resolution, given the geological and operational constraints of the site (Printz et al., 2022).

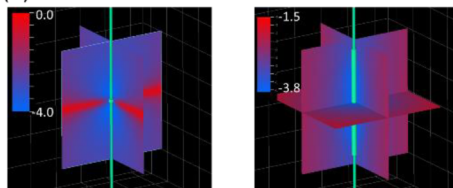
Next, waveform simulations are conducted to model the expected seismic signatures of various failure mechanisms. These synthetic datasets serve as benchmarks for validating the acquisition system and training automated detection algorithms.

A real-time DAS data simulator is then used to emulate field conditions, allowing for end-to-end testing of the acquisition and processing pipeline. This step is critical for verifying system performance before deployment (Mizuno and Le Calvez, 2024b).

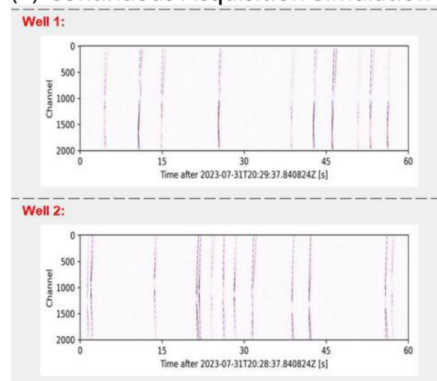
The event detection module integrates traditional geophysical processing with image analysis techniques to identify and classify microseismic events in real time (Drew et al., 2004; Le Calvez, 2007; Mizuno et al., 2021) (Figure 1).

Simulation Technologies

(a) Survey Design

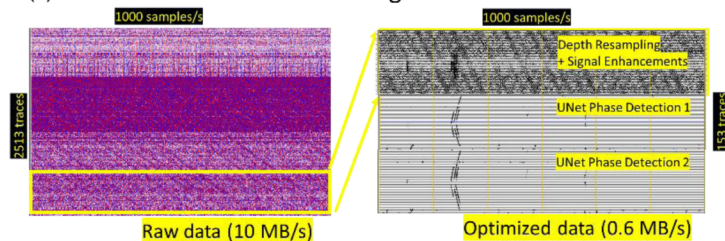


(b) Continuous Acquisition Simulation



Real-time DAS Geophysics

(c) Wellsite Waveform Processing



(d) Automated Event Location

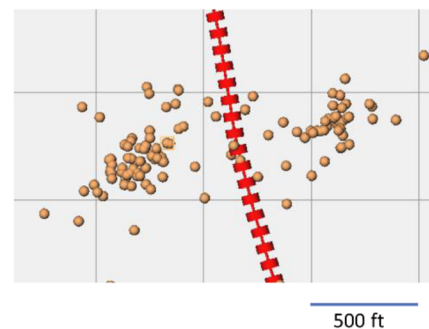


Figure 1—Overview of the foundational technologies for real-time DAS passive seismic monitoring. (a) This panel illustrates a comparison of detectability maps—representing the minimum magnitude threshold across space—between a single DAS receiver (left) and an array of 40 DAS receivers (right). The monitoring well, shown as a green vertical trajectory, includes receiver positions marked by disks. The expanded array significantly enhances event detectability across the monitored volume. (b) This example shows synthetic continuous DAS waveforms generated by a real-time simulation tool. Depending on the magnitude and hypocentral location of each event, some are detectable in both wells, while others are only visible in one, highlighting the spatial sensitivity of the DAS system. (c) On the left, raw continuous seismic data from the DAS interrogator box are displayed. On the right, the processed waveform data at the well site is shown. Processing steps include depth trimming and signal enhancement to facilitate quality control and efficient analysis. Additionally, a "mask" trace may be computed on-site and appended to the data, indicating the likelihood of passive seismic arrivals within the waveform. (d) This panel presents a map view of event locations estimated using a migration-based method that incorporates the mask trace from (c). Since the DAS array is deployed in a single borehole (red trajectory), and only data from the horizontal section are processed (with sensing points shown as red disks), the inversion assumes all events occur at the same depth as the borehole. (Modified from Mizuno and Le Calvez, 2024)

This architecture provides a robust foundation for scalable, real-time seismic monitoring across diverse energy applications.

Monitoring Geometry Optimization

The effectiveness of any passive seismic monitoring system—particularly one based on Distributed Acoustic Sensing (DAS)—is highly dependent on the design of the monitoring geometry. Optimal sensor placement ensures high event detectability, accurate location resolution, and comprehensive spatial coverage of the reservoir or target formation. In DAS systems, where sensing points are distributed along a continuous fiber optic cable, the geometry is influenced by well trajectory, fiber installation method (e.g., wireline or cemented), and the proximity of the fiber to expected seismic sources (Mizuno and Le Calvez, 2023).

The first step in optimizing the monitoring geometry involves constructing a detailed velocity model of the subsurface. This model incorporates geological layering, lithological properties, and geophysical

parameters such as P- and S-wave velocities. Where available, geomechanical data—such as in-situ stress fields, fracture orientations, and rock strength—are integrated to simulate realistic failure scenarios. Attenuation characteristics and ambient noise levels are also considered to assess signal-to-noise ratios and refine sensor placement.

Using this velocity model, synthetic microseismic events are generated at various locations and depths within the reservoir. These events are used to evaluate the detectability and location uncertainty for different fiber configurations. Metrics such as azimuthal coverage, ray path density, and travel time residuals are analyzed to identify blind spots and optimize the array layout.

In multi-well scenarios, the use of cross-well DAS arrays can significantly enhance spatial resolution and event triangulation. The flexibility of DAS allows for rapid reconfiguration and the use of existing infrastructure, making it a cost-effective solution for both temporary and permanent monitoring.

Ultimately, this optimization process ensures that the DAS system is tailored to the specific geological and operational context of the site, maximizing its ability to detect and accurately locate microseismic events in real time (Figure 2).

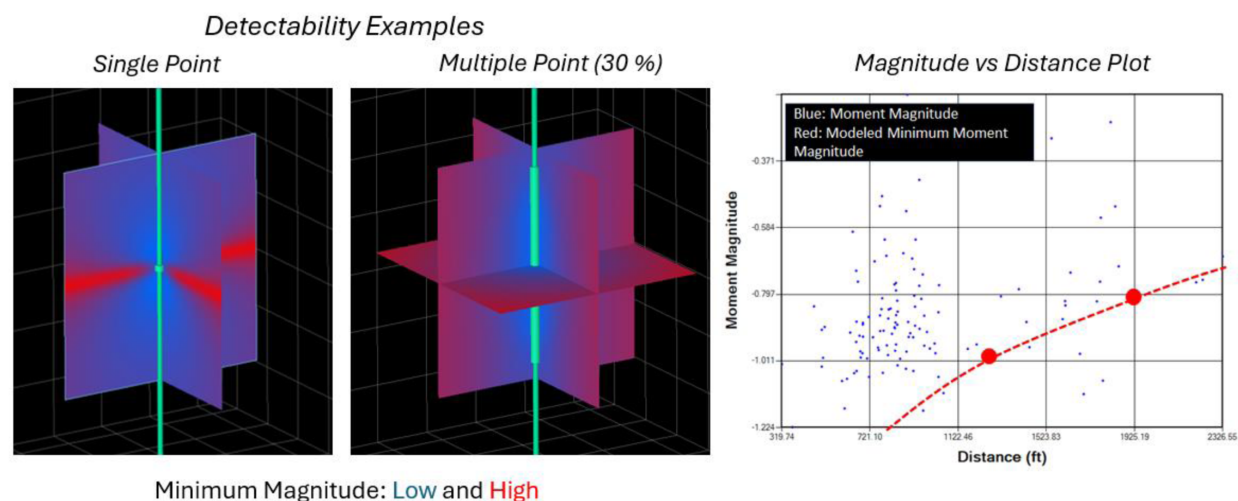


Figure 2—Left: Single 1C measurement (left) displays directionality (slowness scaling of DAS and 1C) as well as distance dependency in terms of detectability. DAS (right) displays only the distance dependency since the directionality is cancelled. Right: minimum magnitude observed is explained by the DAS detectability estimate (Modified from Mizuno and Le Calvez, 2023)

Seismic Waveform Simulation

Seismic waveform simulation plays a critical role in validating the monitoring system design and preparing for real-time interpretation of microseismic events. In the context of Distributed Acoustic Sensing (DAS), waveform modeling is particularly important due to the unique directional sensitivity and spatial sampling characteristics of fiber optic sensors. Simulating realistic waveforms allows for benchmarking the system's performance, training detection algorithms, and improving the interpretability of recorded data.

The simulation process begins with the generation of synthetic microseismic events based on plausible failure mechanisms, such as shear slip, tensile opening, or mixed-mode fracturing. These mechanisms are selected based on the geomechanical properties of the reservoir and the expected stress regime. Each event is characterized by its source parameters, including location, moment tensor, and rupture dynamics.

Using the velocity model developed during the monitoring geometry optimization phase, wave propagation is simulated through the subsurface to the DAS array. The resulting synthetic waveforms are processed to reflect the response of the fiber optic cable, accounting for gauge length, coupling conditions, and directional sensitivity. This step is essential for understanding how different types of events will appear in DAS data and for distinguishing between true seismic signals and noise.

The simulated waveforms are then used to evaluate the system's ability to detect and characterize events under various conditions. This includes assessing the impact of noise, attenuation, and sensor spacing on signal clarity and arrival time accuracy. Additionally, the synthetic data serve as a training set for machine learning or image-based detection algorithms, enabling more robust and automated event identification.

By incorporating waveform simulation into the workflow, operators can anticipate the seismic response of the reservoir, refine their monitoring strategy, and enhance the reliability of real-time seismic interpretation (Figure 3a and Figure 3b).

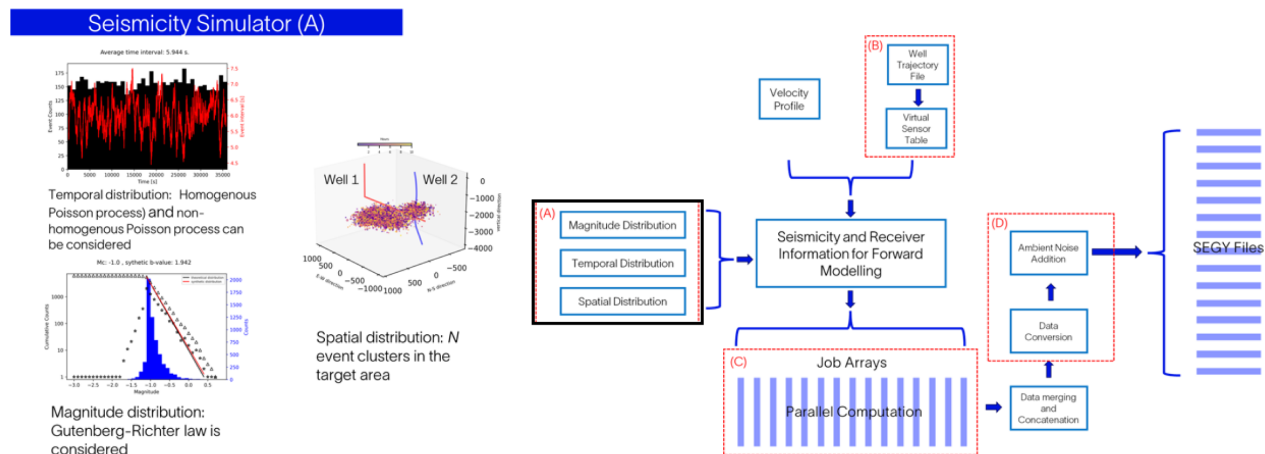


Figure 3a—Overview of the real-time data simulator (Part A) which leverage a statistical model to populate a seismicity catalogue.

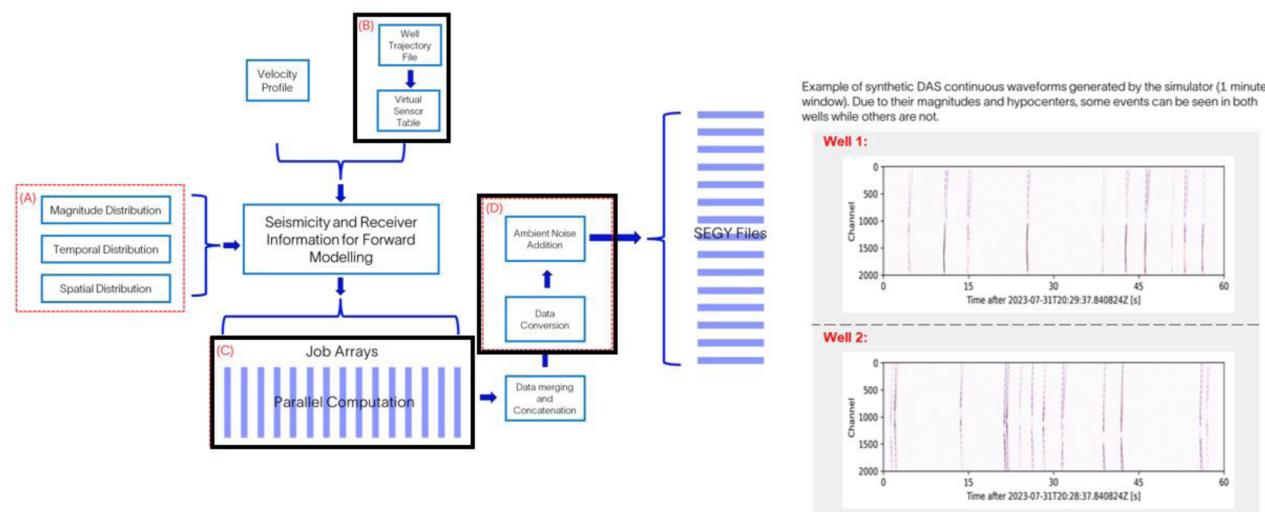


Figure 3b—Overview of the real-time data simulator (Part B, C and D) which is here a ray-based synthetic and DAS response simulator to compute DAS microseismic activity.

Real-Time DAS Data Simulation

To ensure the effectiveness of a DAS-based passive seismic monitoring system before field deployment, it is essential to simulate the entire data acquisition and processing workflow under realistic conditions. Real-time DAS data simulation serves as a critical validation step, enabling operators to test system performance, optimize processing algorithms, and anticipate operational challenges.

The DAS data simulator replicates the behavior of a deployed fiber optic sensing system by generating synthetic seismic data streams in real time. It incorporates the effects of wave propagation through the subsurface, the directional sensitivity of the fiber, and the specific characteristics of the DAS interrogator

unit, such as gauge length, sampling rate, and noise floor. The simulator also models the coupling conditions between the fiber and the surrounding formation or wellbore infrastructure, which can significantly influence signal fidelity. By feeding synthetic microseismic events—generated from waveform simulations—into the simulator, users can evaluate how different event types and source mechanisms are recorded by the DAS system. This allows for the testing of various acquisition geometries, including single-well, cross-well, and surface-deployed fiber configurations. The simulator can also introduce realistic noise sources, such as surface activity, fluid flow, or equipment vibrations, to assess the robustness of the detection algorithms.

One of the key advantages of real-time simulation is the ability to test the full processing pipeline, from raw data acquisition to event detection and characterization, under time constraints that mimic actual field operations. This helps identify potential bottlenecks, validate latency requirements, and fine-tune automated workflows.

Ultimately, the DAS data simulator (Figure 4) provides a safe, cost-effective environment for system calibration and operator training. It ensures that the monitoring system is fully prepared for deployment, reducing uncertainty and enhancing confidence in the system's ability to deliver reliable, real-time seismic insights.

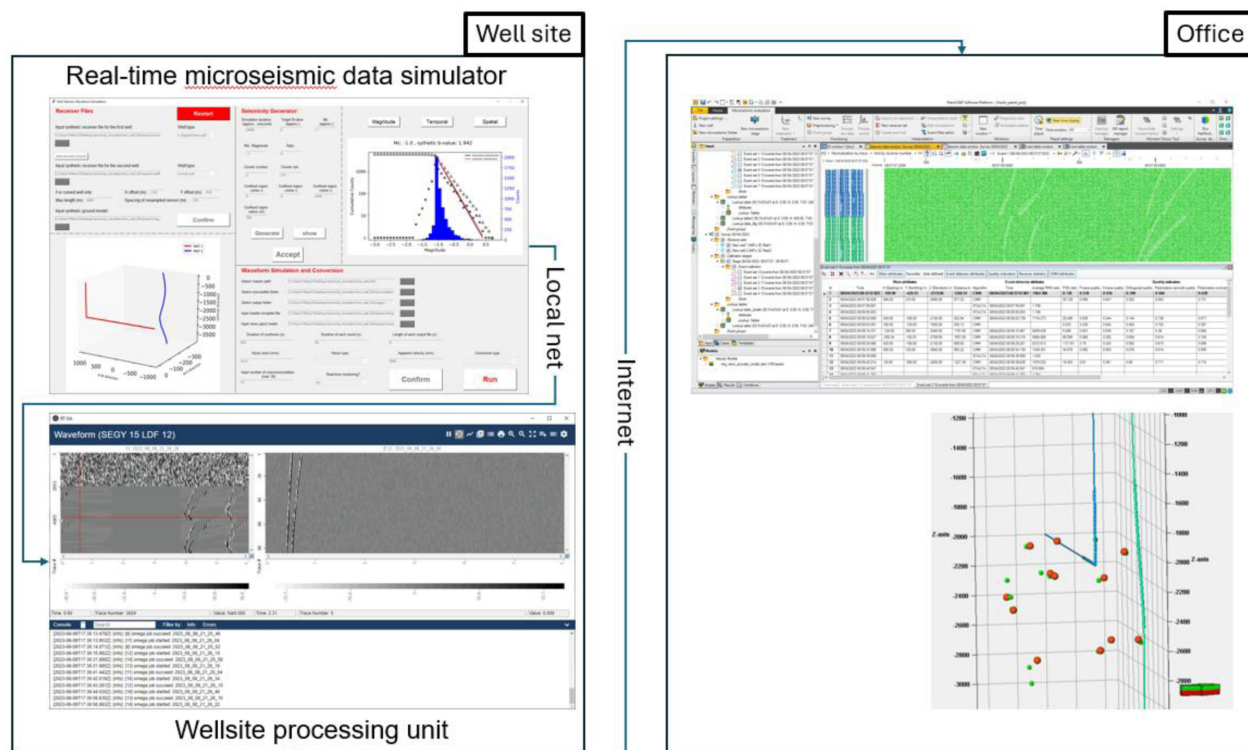


Figure 4—Use of the real-time data simulator to test the entire processing chain.

Event Detection and Processing Framework

Accurate and timely detection of microseismic events is central to the success of any passive seismic monitoring system. In DAS-based systems, the challenge is compounded by the high data volume, directional sensitivity of the fiber, and the often lower signal-to-noise ratio compared to conventional 3C sensors. To address these challenges, we developed a real-time event detection and processing framework that integrates traditional geophysical techniques with advanced image processing methods.

The framework begins with preprocessing steps that enhance signal quality and suppress noise. These include bandpass filtering, amplitude normalization, and adaptive noise suppression tailored to the

characteristics of DAS data. Following preprocessing, the data are transformed into visual representations—such as time-distance plots or spectrograms—that highlight coherent seismic energy across the fiber array.

To automate event identification, we could either follow traditional STA/LTA approach or apply image processing algorithms that mimic the pattern recognition capabilities of expert geophysicists. Techniques such as edge detection, clustering, and morphological filtering are used to isolate seismic arrivals from background noise. These methods are particularly effective in identifying weak or overlapping events that may be missed by conventional amplitude-based detectors.

Once candidate events are identified, arrival times are picked using a combination of cross-correlation and machine learning-assisted algorithms. These picks are then used to estimate event locations and origin times. The system is designed to operate in real time, with low latency, enabling continuous monitoring and rapid response to anomalous activity.

The modular nature of the framework allows it to be deployed on edge devices for local processing or integrated into cloud-based platforms for centralized analysis. This flexibility supports scalable deployment across multiple wells or fields.

By combining geophysical rigor with modern data science techniques, this framework significantly enhances the reliability and automation of DAS-based microseismic monitoring.

Examples of potential improvements

Joint Inversion for Source Characterization

Accurate characterization of microseismic events is essential for understanding subsurface behavior, assessing containment risks, and informing operational decisions (Le Calvez and Mizuno, 2023). Traditional workflows often rely on separate steps for event location and velocity model calibration, typically using active sources or repeated manual adjustments. This approach can be time-consuming, labor-intensive, and prone to inaccuracies—especially in dynamic environments where the velocity field evolves over time due to fluid injection or pressure changes.

To address these limitations, we implement a joint inversion methodology that simultaneously estimates the event hypocenter and updates the velocity model. This integrated approach improves the accuracy of event localization and source mechanism analysis while reducing the need for manual calibration. It is particularly well-suited for DAS-based systems, where dense spatial sampling provides rich waveform data but also introduces complexity in interpretation.

The inversion process begins with initial estimates of the velocity model and event location, derived from arrival time picks and geological constraints. Using an iterative optimization algorithm, the model adjusts both the velocity structure and the event parameters to minimize the misfit between observed and predicted travel times. This dual update process allows the system to adapt to time-varying conditions and heterogeneous formations.

Incorporating DAS-specific considerations—such as directional sensitivity and gauge length effects—into the inversion framework ensures that the results are physically meaningful and tailored to the sensing modality. The method also supports moment tensor inversion, enabling the determination of source mechanisms and fault orientations.

By unifying velocity calibration and event localization, this joint inversion approach enhances the reliability of microseismic interpretation. It provides a more complete picture of subsurface processes, supporting real-time decision-making and long-term monitoring objectives across CCS, hydraulic fracturing, and geothermal applications.

Moment Tensor Inversion

Moment tensor inversion is a critical technique in microseismic monitoring, particularly in the context of hydraulic fracturing, geothermal energy extraction, and carbon sequestration. It involves the mathematical

reconstruction of the seismic source mechanism from recorded waveforms, allowing geoscientists to infer the nature and orientation of subsurface fractures. In microseismic monitoring, where events are typically of low magnitude, moment tensor inversion provides insights into the stress regime and fracture dynamics of the reservoir. The moment tensor itself is a second-order tensor that encapsulates the force system responsible for the seismic event, including shear slip, tensile opening, and volumetric changes. Inversion techniques rely on accurate velocity models and high-quality waveform data from dense sensor arrays to resolve the tensor components. The process typically involves forward modeling of synthetic waveforms and iterative optimization to minimize the misfit between observed and modeled data. Challenges in moment tensor inversion include noise contamination, limited station coverage (also referred to ‘focal sphere coverage’), and non-uniqueness of solutions, which can lead to ambiguities in interpreting the source mechanism. To address these issues, advanced inversion methods incorporate constraints such as known fault orientations or use probabilistic frameworks to quantify uncertainties. The results of moment tensor inversion are often visualized using beachball diagrams, which graphically represent the faulting style—normal, reverse, strike-slip, or complex combinations. These interpretations are crucial for understanding fracture propagation, optimizing stimulation strategies, and assessing the integrity of subsurface formations. As computational capabilities and sensor technologies advance, moment tensor inversion continues to evolve, enabling more precise and real-time characterization of microseismic sources. This enhances the ability of geoscientists and engineers to make informed decisions in subsurface resource management and environmental monitoring.

Discussion

The field applications and workflow components presented in this study highlight the transformative potential of Distributed Acoustic Sensing (DAS) for passive seismic monitoring across a range of energy sectors. Compared to traditional 3C geophone-based systems, DAS offers significant advantages in terms of spatial resolution, deployment flexibility, and cost-effectiveness—particularly when leveraging existing fiber infrastructure. However, these benefits come with unique challenges that must be addressed through tailored processing and interpretation strategies.

One of the key strengths of the proposed workflow is its modularity and adaptability. Each component—from monitoring geometry optimization to joint inversion—can be customized to suit specific geological settings, operational goals, and regulatory requirements. This flexibility makes the workflow applicable not only to carbon storage and hydraulic fracturing, but also to geothermal energy production and long-term monitoring in mature oil and gas fields.

The integration of image processing techniques into the event detection framework represents a notable advancement. By mimicking expert interpretation and enhancing weak signal recognition, this approach improves detection reliability in noisy environments—a common limitation in DAS systems. Similarly, the joint inversion methodology addresses a long-standing challenge in microseismic monitoring: the need for accurate, dynamic velocity models. By coupling velocity updates with event localization, the workflow reduces uncertainty and enhances the interpretability of seismic data.

Despite these advancements, several limitations remain. DAS systems are inherently sensitive to axial strain, which can limit their ability to detect certain wave types or event orientations. Additionally, the high data volume generated by DAS arrays requires robust data management and processing infrastructure, particularly for real-time applications.

Future work should explore the integration of machine learning for adaptive event classification, the use of hybrid DAS–3C systems for improved azimuthal coverage, and the development of standardized protocols for DAS-based MMV programs. These efforts will further solidify DAS as a cornerstone technology for next-generation subsurface monitoring.

Conclusions

This study highlights a transformative approach to passive seismic monitoring through the integration of advanced processing techniques with traditional seismic workflows. By leveraging both 3C and distributed fiber optic sensing technologies, the proposed framework significantly enhances the fidelity and reliability of real-time subsurface monitoring. The ability to simulate the complete seismic workflow—including forward modeling, waveform synthesis, and data transmission—prior to deployment enables proactive system optimization and improved event detection, particularly in geologically complex or noisy environments. Moreover, the implementation of a scalable Edge-Cloud architecture ensures that the system can meet the demands of high-throughput, real-time data processing across diverse operational scales. These innovations collectively offer a robust, future-ready solution for geoscientists and industry leaders seeking to improve reservoir characterization, operational safety, and decision-making in dynamic field environments.

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